

A PROJECT TITLE

ON

< Parking Demand Prediction Using Elastic Net Regularized Regression >

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE ENGINEERING

in

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by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: “Parking Demand Prediction Using Elastic Net–Regularized Regression”

1. ABSTRACT:

Finding a parking space in busy urban areas has become a common challenge due to the growing number of vehicles and limited parking facilities. Drivers often spend considerable time searching for available parking spots, which contributes to traffic congestion, fuel wastage, and environmental pollution. To address this issue, predicting parking demand in advance can help authorities manage parking resources more efficiently and improve the overall transportation experience. This project, “Parking Demand Prediction Using Elastic Net Regularized Regression,” aims to develop a machine learning model that can accurately forecast parking demand based on historical data and other relevant factors.

The proposed system uses data such as parking occupancy records, time and date information, weather conditions, holidays, and local events that may influence parking usage. The collected data is first cleaned and processed to remove inconsistencies and improve its quality. Feature engineering and normalization techniques are then applied to prepare the data for analysis. The prediction model is built using elastic net regularized regression, a powerful machine learning algorithm that combines the strengths of both Lasso and Ridge regression. This approach helps in selecting the most important features

while reducing the impact of overfitting and highly correlated variables.

After training, the model can predict future parking demand for different locations and time periods. These predictions can support parking operators, city planners, and transportation authorities in making better decisions regarding parking allocation and traffic management. The effectiveness of the model is evaluated using performance measures such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R-squared (R^2).

Overall, this project demonstrates how machine learning can be used to solve real-world urban challenges. By providing accurate parking demand forecasts, the proposed system contributes to smarter parking management, reduced congestion, and improved urban mobility, making cities more efficient and convenient for both drivers and administrators

2. INTRODUCTION:

Parking has become a major challenge in busy urban areas because of the increasing number of vehicles and limited parking spaces. Drivers often spend extra time searching for available parking, which can lead to traffic congestion, fuel wastage, and pollution. Predicting parking demand in advance can help manage parking spaces more effectively.

This project, **“Parking Demand Prediction Using Elastic Net–Regularized Regression,”** uses machine learning to predict parking demand based on factors such as parking occupancy, time, weather conditions, holidays, and local events. Elastic Net Regression is used because it combines the advantages of Lasso and Ridge Regression and helps reduce overfitting. The predicted results can support better parking management and contribute to smoother and more efficient urban transportation.

3. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows / macOS / Linux
- **Programming Language:** Python
- **Platform:** Jupyter Notebook / Google Colab

- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- **Algorithm:** Elastic Net Regression
- **Dataset:** Parking demand/occupancy dataset
- **IDE:** VS Code / Jupyter Notebook

Signature of the Guide Course Instructor